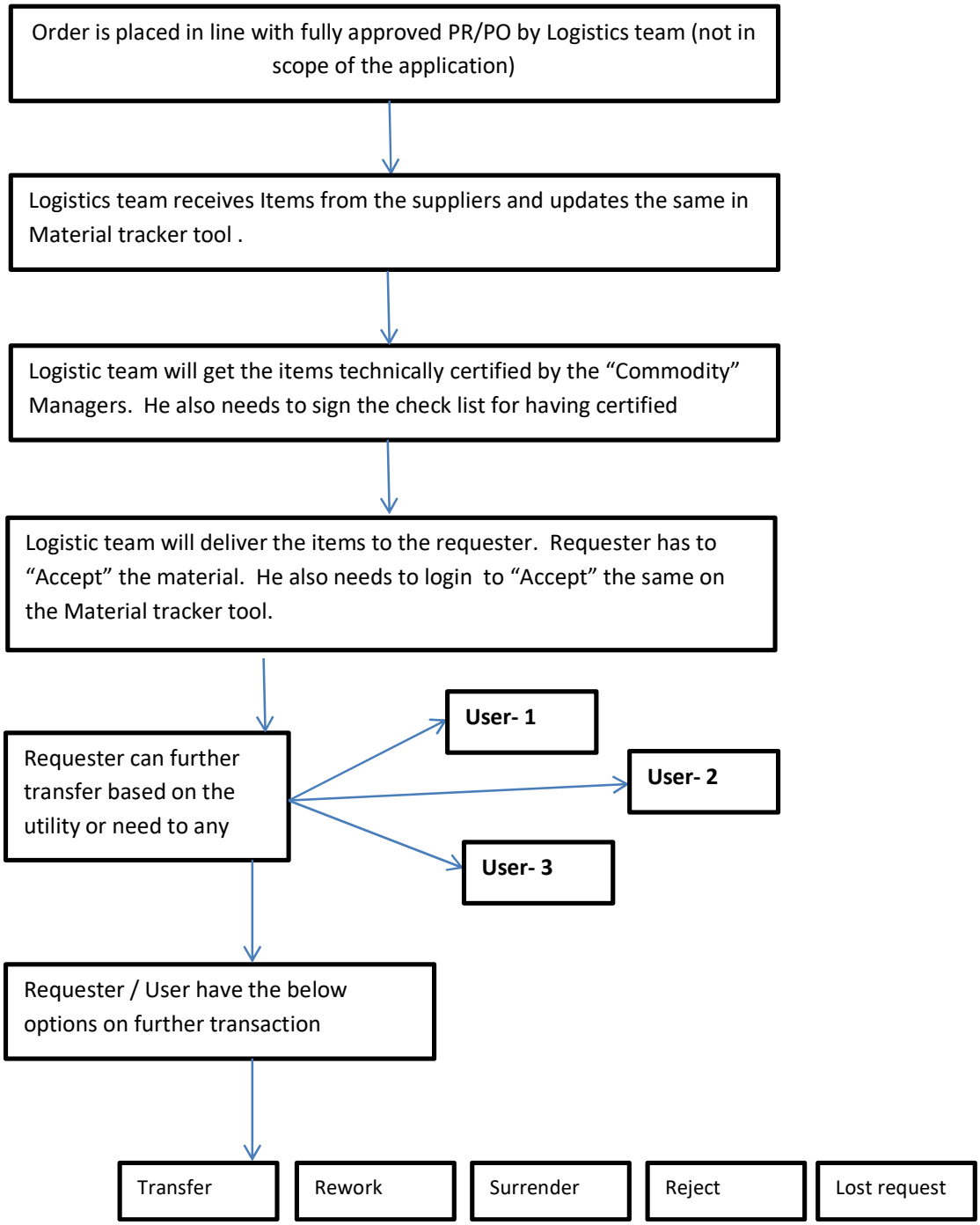
Order is placed in line with fully approved PR/PO by Logistics team (not in scope of the application)
Logistics team receives Items from the suppliers and updates the same in Material tracker tool .
Logistic team will get the items technically certified by the "Commodity" Managers. He also needs to sign the check list for having certified
Logistic team will deliver the items to the requester. Requester has to "Accept" the material. He also needs to login to "Accept" the same on the Material tracker tool.
Requester can further transfer based on the utility or need to any
User- 1
User- 2
User- 3
Requester / User have the below options on further transaction
Transfer
Rework
Surrender
Reject
Lost request

Material Tracker Application – Software Requirements Specification (SRS)

This document outlines the functional and non-functional requirements for the Material Tracker Application, combining details from the provided user text and PDF documents (pp. 1-6). It is intended for the development team as a single source of truth.

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1. Introduction

1.1 Purpose

The purpose of this software requirements specification (SRS) is to provide a detailed blueprint for the development of a web-based **Material (Items) Tracking Application**. This application will manage item tracking within an organization, from initial receipt to transfer, rework, surrender, rejection, or loss.

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1.2 Scope

The system will manage user access, item master data, and track material movement and status. It will use PHP 8.x (without Laravel), MySQL, Bootstrap 5, XAMPP, jQuery-AJAX, and OpenLDAP. The system will *not* include the creation or management of indents for items.

1.3 Definitions and Acronyms

- **PR/PO:** Purchase Requisition/Purchase Order
- **QA:** Quality Assurance
- **SRS:** Software Requirements Specification
- **TP Vision:** A team/entity within the organisation that can receive transfers of items (p. 2).

2. Overall Description

2.1 Product Perspective

The system is a new, standalone web application that integrates with an existing OpenLDAP service for user authentication. It will manage inventory and stock data.

2.2 User Classes and Characteristics

- **System Users (Admin/User):** Employees with `emp_ID` in the format "AAxxxx" or "YYxxxx" (e.g., AA0001) with a Name, Email ID, Start/End Dates, Designation, and Role (p. 1).

- **Logistics-Manager (Admin Role):** Has all admin powers. Responsible for creating PO/PR records and accepting materials into the main stores after QA is done by the Commodity Manager. Manages inventory accuracy.
- **Commodity Manager:** Responsible for QA of incoming items.
- **Requester/Item Owner (User Role):** The end-user who receives the item, acknowledges acceptance, and manages its status (transfer, rework, surrender, reject, lost) (pp. 1-2).
- **"Check Point":** An entity responsible for tracking and managing lost items (p. 4).

2.3 Operating Environment

The application will operate using PHP 8.x, MySQL as the database, and utilize XAMPP, Bootstrap 5 for UI, jQuery-AJAX for front-end interactivity, and integrate with OpenLDAP.

3. Specific Requirements

3.1 Functional Requirements

3.1.1 User Management

- The system shall implement **Role Based Access** (Logistics_Manager, User, Commodity_Manager) control (p. 1).
- User access shall be validated via **LDAP** (p. 1). Provision to be made for using the OpenLDAP and AD , for the Login Authentication. Admin will change the settings in the DB table.
- The system shall support a simple user master for managing employee details (ID, Name, Email, Role, etc.).

3.1.2 Item Master Management (Logistics-Manager)

- The system shall store item details with a numeric `Item_ID` (e.g., 000001) (p. 1).
- The system shall allow the Logistics Manager (Admin) to add item descriptions, category, material_type, and active/inactive status.
- The system shall track quantity (`Qty`) as Number or Batch (p. 1).
- The system shall store a sequential 6-digit `Item_code` (e.g., 000001) as a primary key for each physical item.
- The system shall record `Material_Type` (e.g., Consumable, Non-Consumable).

3.1.3 Item Inward/Acceptance (Logistics-Manager, Commodity Manager, User)

- The Logistics Team shall update incoming material details in the system in line with approved PR/PO (p. 1).
- The system shall capture Serial Numbers and PO # when items are accepted in stores (p. 1).
- The Commodity Manager shall perform QA using a standard checklist, and the QA Cert No shall be recorded (p. 1).
- The system shall track item status: Accepted, Rejected, To-Rework, In-QA, Lost, Lost-but-found (p. 1).
- Upon receipt, the Requester/User must acknowledge and accept items in the system (p. 1).
- The Requester shall be able to upload an image of the item's condition (p. 1).
- Logistics will only hand over material once the item is accepted by the Requester (p. 1).

3.1.4 Item Management Actions (Requester/Item Owner)

After acceptance, the user can initiate the following requests via the system (p. 2):

- **Transfer:**
  - Transfer items to other TP Vision team members (p. 2).
  - Transferee must accept the item; liability remains with the transferor until accepted (p. 2).
  - User must select item number, select "Transfer" acknowledgment, select "Transfer to" name, enter quantity, provide mandatory remarks, and optionally upload a file (p. 3).
- **Rework:** Send an item for repair or replacement (p. 4).
- **Surrender:** Surrender an item when no longer needed (p. 4).
- **Reject:** Reject an item if damaged or not as per requirement (p. 4).
  - For Rework/Surrender/Reject, the user must select the item, the appropriate acknowledgment option, select the "transfer to" (Commodity) owner, enter quantity, provide mandatory remarks, and optionally upload a file (p. 4).
- **Lost:**
  - Item owner must raise a request for a lost item if not found (p. 4).

- Mandatory fields include item number, "Lost" acknowledgment, quantity, remarks, lost date & time, and optional file upload (p. 5).
- The user must still own the liability for the lost item even after raising the request (p. 5).
- If a lost item is found by the requester, they can update the found quantity (p. 4).
- If found by the "Check Point," it is transferred to the requester, who must accept it (p. 4).

- **Transfer Lost Qty:**

- If the item remains lost after checking with the "Check Point," the requester raises a "Transfer Lost quantity" request (p. 6).
- This requires budget approver approval and results in the cost being deducted from the final settlement (p. 6).

3.2 Other Requirements

Email notifications: to go for every actions by User or Admins. Logs to capture such notifications.

Laravel framework: **Not** to be used as the servers may have some restrictions.

MVC Patterns : is required and to be used while building the application.

Reports: Use any Bootstrap dataTable features or so for the templates to display inventory & status, etc.

Audit Logs : For each action by the Users, there should be a log captured. Including the LDAP access by the Admin.

Page Layouts:
All pages should have Header, Footer, Sidebar menu. Sidebar should have menus grouped for Operations, Reports, Feedback & User Manual.
Header to show the Logo on top left corner, title “Material Tracker” in the center, UserName with email Id & current date will be on top right corner.
Footer will have one line display of Copyrights. Use very professional Corporate Blue Color for the borders and all shades. Header bar and Sidebar should be pinned to the top of all pages despite the movement of contents vertically or horizontally using the spacebars.

Sidebar Menus will display menus based on the Roles.

User can see the stocks & past history for his emp_id only.

Logistics_Manager can do all the functions of inward, transfer, Lost & Found, accepting Surrendered items, Managing the Rework Items, etc

Commodity_Manager will be able to do "Transfer Lost Qty" process as shown below:

Transfer Lost Qty : if the requester and "check point" not able to find the lost quantity, the requester as to raise **Transfer Lost quantity** and the same will be approved the by respective budget approver with the option given below.

(i) Item lost will be approved as way-off of lost quantity subject to the cost of the Lost items.
(ii) Item lost cost will be deducted in the Final Settlement.

- **Performance:** The system should have fast response times using jQuery-AJAX.
- **Security:** Access is based on LDAP authentication and role-based permissions.
- **Usability:** The UI should be responsive using Bootstrap 5.
- **Reliability:** The system should maintain accurate inventory records and track liabilities.